



# Object Detection

AZURE CUSTOM VISION

# Objective

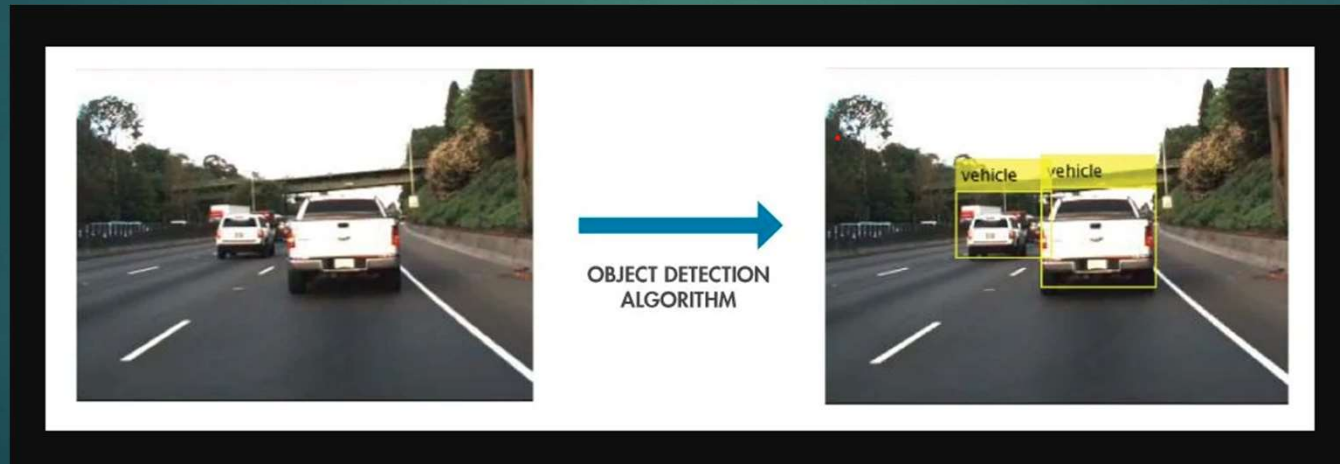
- ▶ Understanding how to develop an API endpoint using Microsoft Azure Custom Vision for object detection.

# Domain

- ▶ Humans can easily identify and detect object present in an image. By leveraging vision capabilities present using vision models, we can automate this process.
- ▶ The real-world applications of object detection can be seen in many crucial areas of our lives, such as medical imaging, video tracking, movement detection, facial recognition, and object recognition, even in autonomous vehicles.

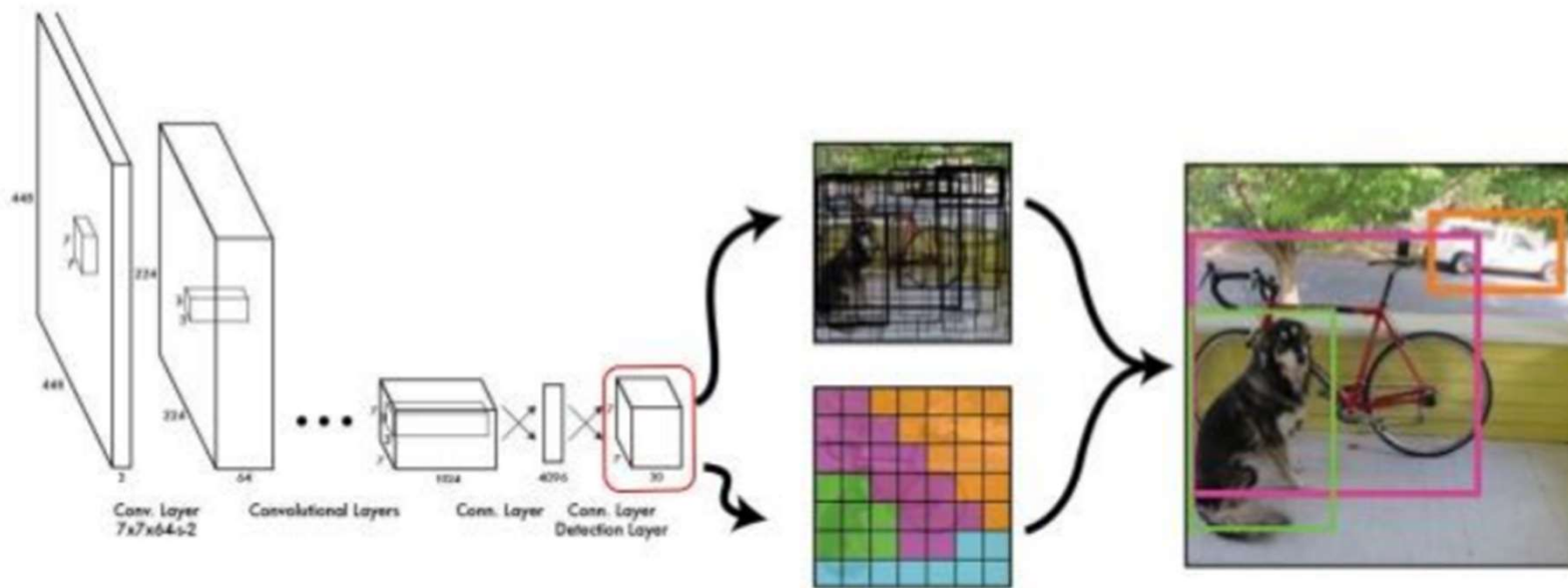
# Object Detection

- ▶ Object detection is a computer vision technique that aims to locate and identify objects within an image or video. Object detection also allows the detection of objects or targets from a visual feed in real time. The process involves steps designed to recognize and label the distinct entities in the graphical data.



# Algorithm Architecture (Basic)

## YOLO: You Only Look Once



# Custom Vision

- ▶ Azure AI Custom Vision is an image recognition service that lets you build, deploy, and improve your own object detection and classification models.
- ▶ The Custom Vision service uses a machine learning algorithm to analyze images for custom features.
- ▶ Custom Vision functionality can be divided into two features. Classification applies one or more labels to an entire image. Detection is similar, but it returns the coordinates in the image where the applied labels are found.

# Prerequisites

- ▶ Azure subscription or create a free account.
- ▶ Need to create a custom vision resource in <https://portal.azure.com>
- ▶ A set of images or sample images downloaded from internet.
- ▶ Supported Web Browser (Chrome, Bing)

# Create Resource

Home > Create a resource > Marketplace > Custom Vision >

## Create Custom Vision ...

Create options \*

☒ Both  
☐ Prediction  
☐ Training

Project Details

Subscription \* ⓘ

Resource group \* ⓘ

[Create new](#)

Instance Details

A training resource and a prediction resource will be created in same region.

Region ⓘ

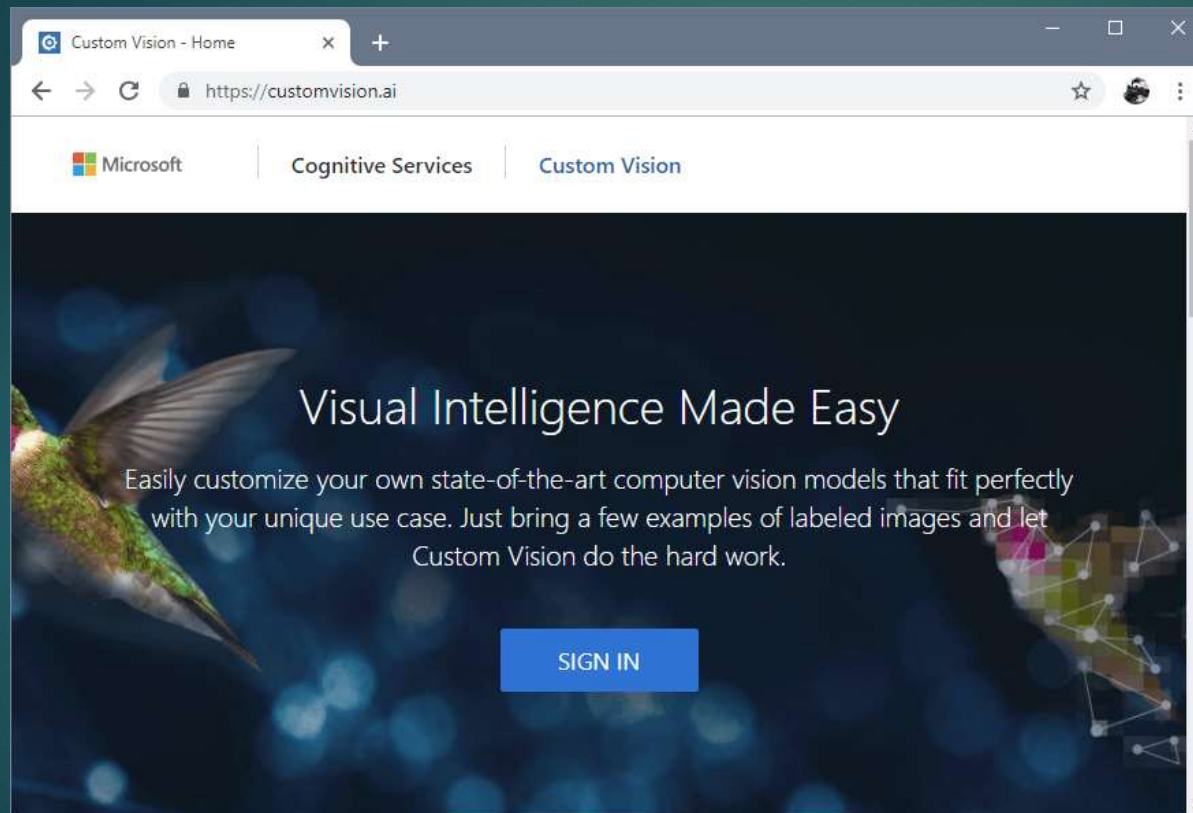
Name \* ⓘ

Training Resource

[Previous](#) [Next](#) [Review + create](#)



# Create a project



## Create new project



Name\*

Enter project name

Description

Enter project description

Resource

[create new](#)

[Manage Resource Permissions](#)

Project Types ⓘ

- ☐ Classification
- ☒ Object Detection

Domains:

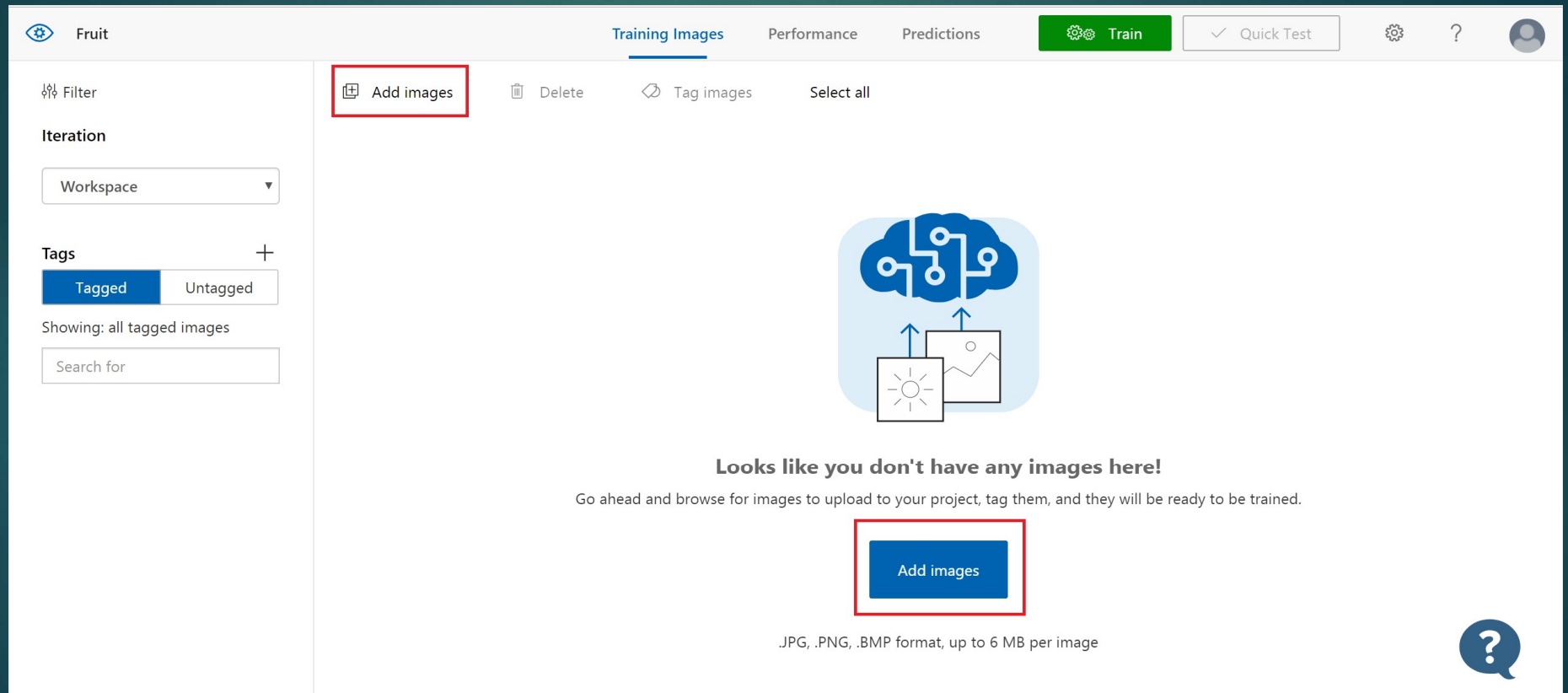
- ☒ General [A1]
- ☐ General
- ☐ Logo
- ☐ Products on Shelves
- ☐ General (compact) [S1]
- ☐ General (compact)

Pick the domain closest to your scenario. Compact domains are lightweight models that can be exported to iOS/Android and other platforms. [Learn More](#)

Cancel

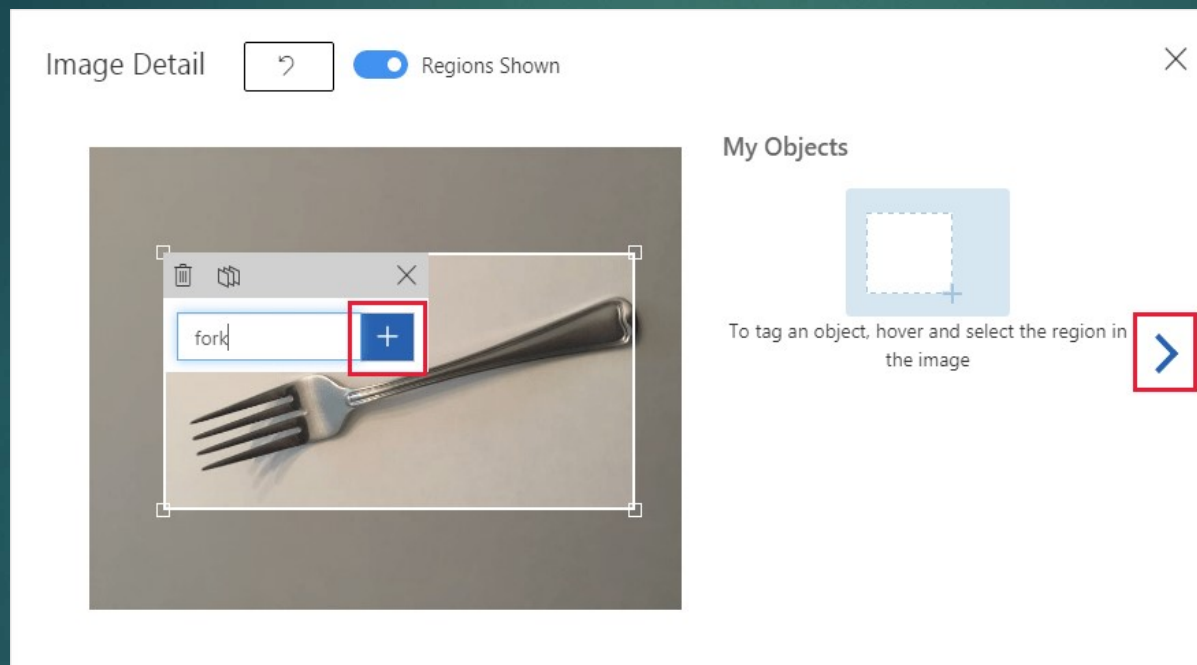
Create project

# Upload and tag images



The screenshot displays the Fruit AI web interface. At the top, a navigation bar includes the 'Fruit' logo, tabs for 'Training Images', 'Performance', and 'Predictions', and buttons for 'Train' (green) and 'Quick Test' (grey). A sidebar on the left contains a 'Filter' section with an 'Iteration' dropdown set to 'Workspace', a 'Tags' section with 'Tagged' and 'Untagged' buttons, and a search bar. The main content area, under the 'Training Images' tab, features a toolbar with 'Add images', 'Delete', 'Tag images', and 'Select all' buttons. The 'Add images' button is highlighted with a red box. Below the toolbar is a large blue icon representing a cloud with circuitry and two image thumbnails. A message states: 'Looks like you don't have any images here! Go ahead and browse for images to upload to your project, tag them, and they will be ready to be trained.' A large blue 'Add images' button is centered below this message and is also highlighted with a red box. At the bottom, text specifies supported formats: '.JPG, .PNG, .BMP format, up to 6 MB per image'. A help icon (question mark in a speech bubble) is in the bottom right corner.

# Tagging



# Train images

**Fruit Demo**

Training Images   **Performance**   Predictions   **Train**   Quick Test

Iterations

Probability Threshold: 50% ⓘ

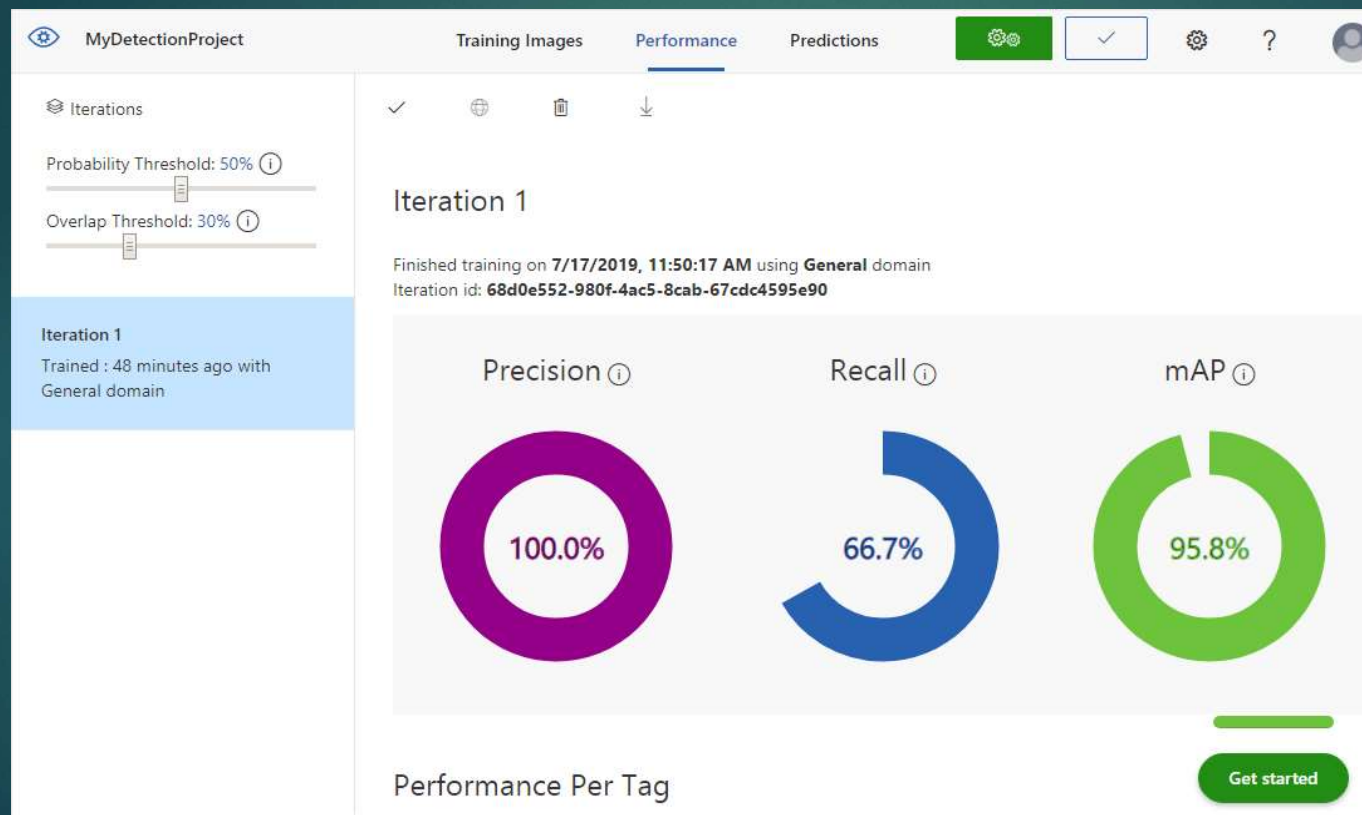
**Iteration 6**  
Training...

Iteration 5  
Trained : moments ago  
with General (compact)  
domain

Delete   Export

**Iteration 6**  
Training...  
Last checked: **7/23/2018, 5:36:49 PM**

# Evaluation



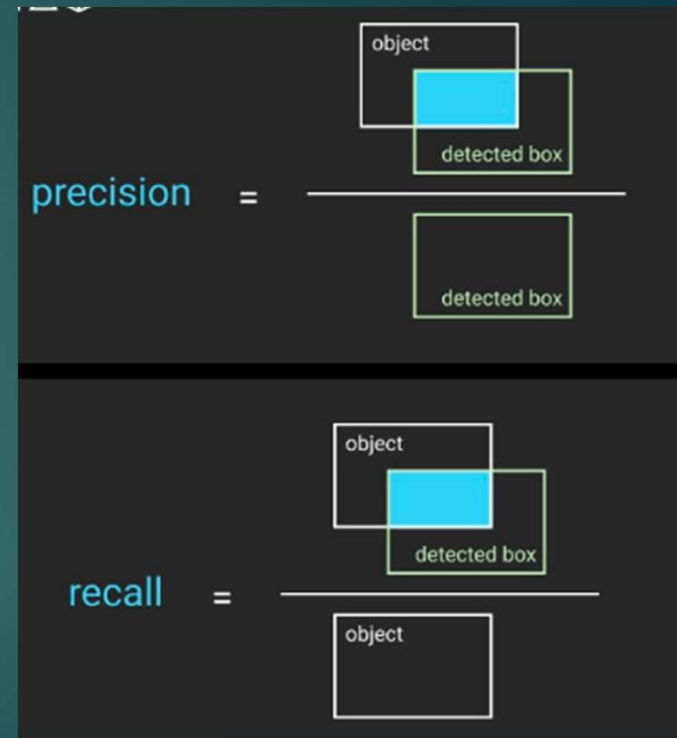
# Metrics

- Precision = Precision measures the accuracy of positive predictions.

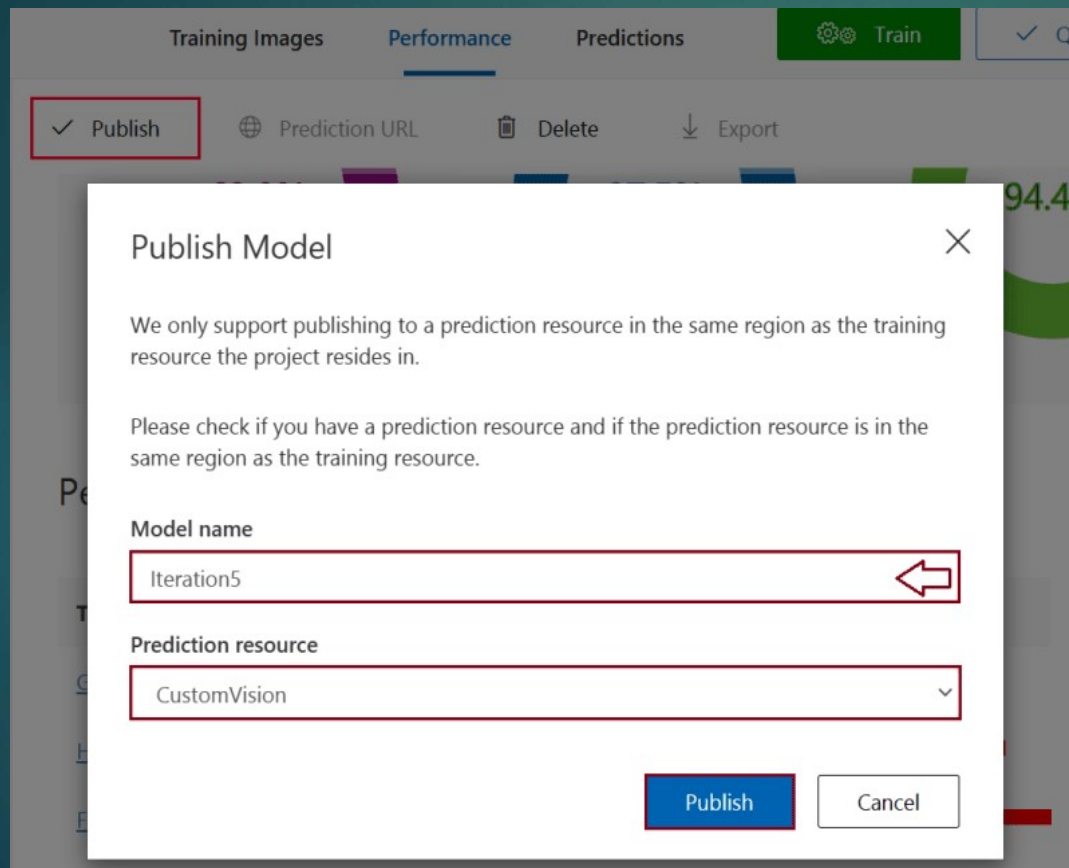
$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

- Recall = Measures the ability to identify all relevant events.  $\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$

- Mean Average Precision (mAP) provides a comprehensive evaluation, especially in multi-class object detection tasks.



# Publishing



Training Images Performance Predictions Train

✓ Publish Prediction URL Delete Export

**Publish Model** ✕

We only support publishing to a prediction resource in the same region as the training resource the project resides in.

Please check if you have a prediction resource and if the prediction resource is in the same region as the training resource.

**Model name**

Iteration5

**Prediction resource**

CustomVision

Publish Cancel



Training Images

Performance

Predictions

Train

Unpublish

Prediction URL

Delete

Export

## How to use the Prediction API

If you have an image URL:

`https://eastus.api.cognitive.microsoft.com/customvision/v3.0/Prediction/`

Set **Prediction-Key** Header to :

Set **Content-Type** Header to : `application/json`

Set Body to : `{"Url": "https://example.com/image.png"}`

If you have an image file:

`https://eastus.api.cognitive.microsoft.com/customvision/v3.0/Prediction/`

Set **Prediction-Key** Header to :

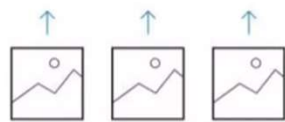
Set **Content-Type** Header to : `application/octet-stream`

Set Body to : `<image file>`

Got it!

# Offline Model

- ▶ Export the trained model for Android,iOs, Docker image etc.
- ▶ The same can be achieved to Azure Python SDK
- ▶ Steps:
  - ❑ Install the necessary packages.
  - ❑ Load the appropriate key values.
  - ❑ Perform the prediction



Upload Images



Train



Evaluate

# Important tips

- ▶ Minimum images per tag is 15.
- ▶ Object should be the focus of the picture.
- ▶ Diverse images for better training and recall value.
- ▶ Image size should not be more than 6MB.
- ▶ The image aspect ratio should not be larger than 25:1.

# References

- ▶ <https://learn.microsoft.com/en-us/azure/ai-services/custom-vision-service/getting-started-build-a-classifier>
- ▶ <https://learn.microsoft.com/en-us/azure/ai-services/custom-vision-service/concepts/compare-alternatives>
- ▶ [https://labelyourdata.com/articles/object-detection-metrics#:~:text=The%20performance%20of%20an%20object,mAP\)%200across%20different%20object%20categories.](https://labelyourdata.com/articles/object-detection-metrics#:~:text=The%20performance%20of%20an%20object,mAP)%200across%20different%20object%20categories.)



Thank you.